



Intelligent Document Q&A System Using RAG

CME4418 — Introduction to Artificial Intelligence

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The Problem: Information Overload & Unreliable AI

Exponential Document Growth

Academic papers, technical reports, and manuals are multiplying faster than any human can read — making targeted information extraction increasingly difficult.

Search Returns Documents, Not Answers

Traditional keyword search locates files but leaves users to read and interpret — it doesn't answer the actual question.

LLMs Hallucinate

Large language models are powerful reasoners, but they fabricate facts when their training data lacks the required knowledge — an unacceptable risk for technical use.

The Gap We Bridge

We need a system that combines **retrieval accuracy** with **LLM reasoning** — grounding every answer in real source text.

What is Retrieval-Augmented Generation?

RAG, introduced by **Lewis et al. (2020)**, is a hybrid architecture that enhances LLMs by pairing them with a live document retrieval system.

Instead of relying solely on static training data, the model **retrieves relevant passages** from an indexed document corpus at query time — then uses those passages as grounded context to generate its answer.

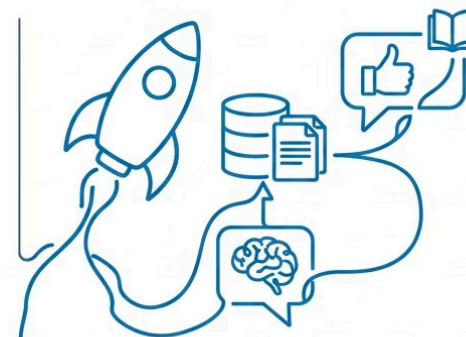
The result: **fewer hallucinations, higher factual accuracy, and traceable sources.**



Traditional Search:
documents only,
no answers.

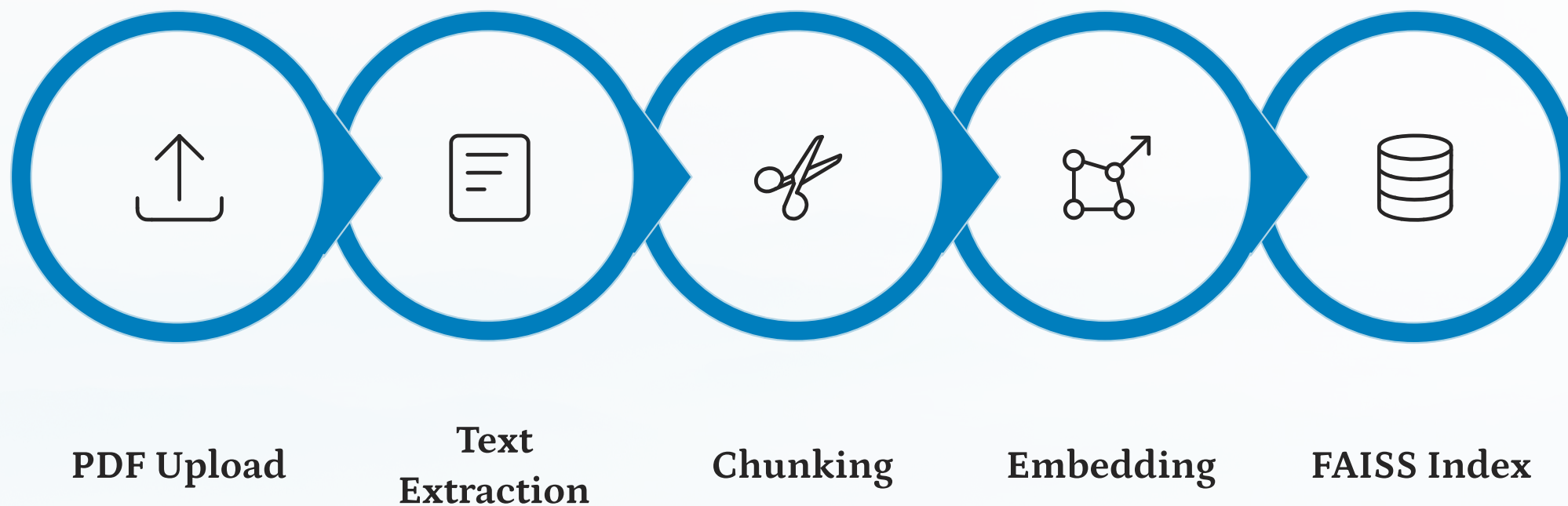


Vanilla LLM:
fluent text,
may
hallucinate.



**RAG: grounded,
accurate
answers, with
source
citations.**

A Two-Phase Pipeline



The indexing phase runs once per document; the querying phase executes in real time on every user question — delivering answers in seconds with full traceability back to the source passages.



Tools & Technologies



Backend & Frontend

Python + FastAPI powers the REST API layer; **React.js** delivers a responsive, intuitive user interface.



Parsing & Chunking

PyMuPDF extracts text from PDFs; **LangChain TextSplitter** segments it into 500-character chunks with 50-character overlap.



Embeddings & Vector Store

sentence-transformers (all-MiniLM-L6-v2) generates dense embeddings; **FAISS** indexes and retrieves the top-5 most relevant chunks.



LLM & Deployment

Anthropic Claude API (claude-haiku-4-5) generates final answers; the entire stack is containerized with **Docker** for reproducible deployment.

Demo & Key Features



Instant PDF Querying

Upload any academic PDF and ask questions in plain natural language — answers appear in seconds.



Source Attribution

Every answer includes the **exact passages and page numbers** used — full transparency, zero guesswork.



Multi-Document Support

Query across **multiple uploaded documents** simultaneously — ideal for literature reviews and cross-referencing reports.



Verifiable & Trustworthy

Users can inspect the retrieved evidence behind every response, making the system audit-ready for academic and professional use.



AI ASSISTANT



What are the benefits of mindful meditation for mental health?

Mindful meditation offers significant mental health benefits, including stress reduction, improved focus, and emotional regulation. It involves focusing on the present moment and accepting thoughts without judgment.



SOURCE PASSAGE

1.mindfulness-based stress reduction (MBSR) program can significantly **lower levels of stress and anxiety** in participants [Ref A].
2.studies indicate **improved attention control and concentration** following regular meditation practice [Ref B].

Ask another question...

